Structure Factor report

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In this report are the main results of the structure factor analysis of two experiments with almost the same systems. The first experiment has a time span of 10000τ with systems of $\phi \in \{0.01, 0.05\}$, and the second experiment has a time span of 500500τ with systems of $\phi \in \{0.01, 0.04\}$. A fit with exponential and Lorentzian functions was performed at the last time instant to see what type of information could be interpreted. The main result is the definition of three spatial zones. One in the order of magnitude of the patchy particles, a second one with an order of magnitude between the box length and the patchy particles' size, and a third one in the order of magnitude of the box length. Finally, I think that simulations with a bigger box size can be usefull to get a better resolution for the third spatial zone.

Results Experiment 1

An exponential fit, $f(|\vec{q}|) = \dfrac{A}{|\vec{q}|^{\alpha}}$, is performed to the structure factor of the systems shown in the following figure **??**.

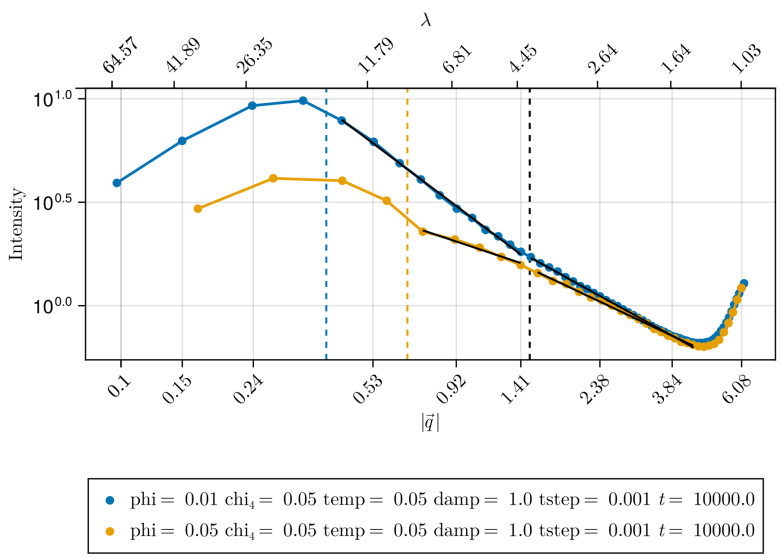


Figure 1: Structure factor of two systems with different packing fraction at the same time instant.

The result of the parameters are shown in the following table **??**. The systems 1 and 2 are represented with blue and yellow colours repectively. Finally, zone 1 is defined as $\lambda_1 \in [7/2\sigma_p, 21/2\sigma_p]$ and zone two is defined as $\lambda_2 \in [21/2\sigma_p, L/4]$.

Discussion

For me it is not clear the interpretation of α. Acoording to a quick[1]

[1] And not to professional. (I ask DeepSeek.)

System	Zone	A	α
1	1	0.925179	2.486543
2	1	0.794425	2.08354
1	2	1.264535	2.731100
2	2	0.561453	1.948924

Table 1: Results of the fit.

research, the convention of an exponential fit is to get the *fractal mass dimension*[2]. In this fit, when $\alpha \approx 2.52$ it is said that the network reached a percolation state. However, there are some subtleties in the spatial domain. Because exponenets near that value are in the order of the patchy particle size, not in the order of box length.

[2] Is a parameter that describes the scaling of mass with radius in a fractal structure.

On the other hand, it is intereset to see that the structure factor can be analyzed in three domains.

Results Experiment 2

In this experiment, the time domain of the simulation was increased. As a consequence, the definition of the spatial zones was done again. The first zone definition remains the same, $\lambda_1 \in [7/2\sigma_p, 21/2\sigma_p]$. In contrast, the upper bound of the second zone was re-defined, $\lambda_2 \in [21/2\sigma_p, 49/2\sigma_p]$. Finally, a third zone was defined, $\lambda_3 \in [49/2\sigma_p, L]$.

For the first and second zone, an exponential fit,$f(|\vec{q}|) = \frac{A}{|\vec{q}|^{\alpha}}$, is performed to the structure factor. Meanwhile the fit on the third zone was performed using a Lorentzian function $f_3(|\vec{q}|) = \frac{S(0)}{1+\xi^2|\vec{q}|^2}$.

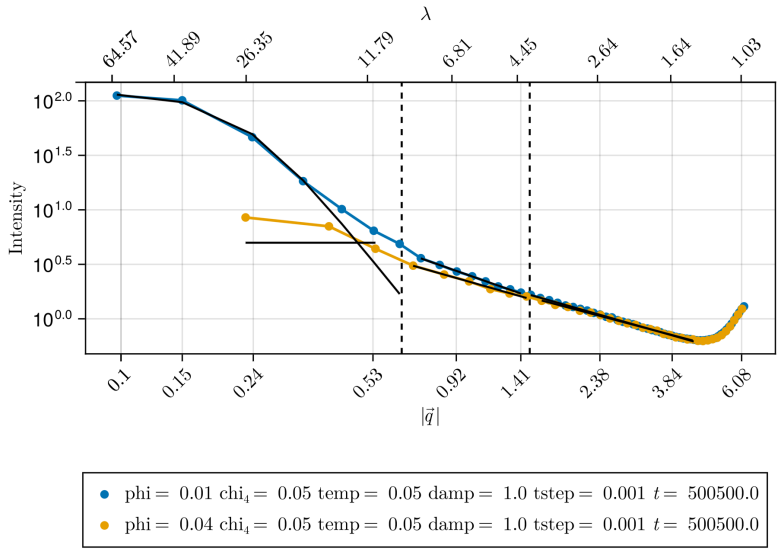


Figure 2: Structure factor of two systems with different packing fraction at the same time instant.

The result of the parameters are shown in the following table **??**.

System	Zone	A	α
1	1	0.910700	2.400446
2	1	0.845499	2.214082
1	2	1.121956	2.196358
2	2	0.907001	1.948924

Table 2: Results of the fit.

System	Zone	S(0)	β	ξ
1	3	118.434653	4.000113	4.557270
2	3	4.992773	37.440534	0.4000000

Table 3: Results of the fit.

Discussion

I'm still having trouble in the interpretation of the exponenetial fit. On the other hand, the fit fit the Lorentzian function is due to the Ornstein-Zernike standard. Where we expect that $\beta \approx 2$, because at $\beta = 2$, the value of ξ is the correlation length and $S(0)$ is related with the density, thermal energy and isothermal compressibillity[3], $S(0) = \rho k_B T \kappa_T$.

[3] $\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial P}\right)_T$

Characteristic distances in the system and the domains for the fit

The distance range for the pair-wise interaction is defined by the distance at minimum energy, to the cutoff, $r_b \in [\sigma_p, 3/2\sigma_p]$. Then, considering that the center of all particles, central and patch, lies on the same line the distances between the center of the central particles is given by $r_{\text{int}} = 2\sigma_p + r_b$. If we evaluate r_b at the limits, the interaction distance is $r_{\text{int}} \in [1.2, 1.4]$. Finally, if we increase the amount of patchy particles in pair-wise interaction, the distance between central particles is given by

$$d_{\text{int}} = (N-1)\left(2\sigma_p + r_b\right).$$

Zone definition of Experiment 1

The result fo this analysis is used to defined the region at small distances. More specifically, the lower limit of this region is set at $N = 2$ and $r_b = 3/2\sigma_p$ and the upper limit is set to $N = 4$ and $r_b = 3/2\sigma_p$. Leading to the following spatial domain,

$$\lambda_1 \in \left[7/2\sigma_p, 21/2\sigma_p\right].$$

Then, we define a second spatial domain, which it's upper limit is set the order of magnitud of the total length of the simulation box, as follows,

$$\lambda_2 \in \left[21/2\sigma_p, L/4\right].$$

Zone definition of Experiment 2

The lower limit of the first zone remains the same as the previous experiment,

$$\lambda_1 \in \left[7/2\sigma_p, 21/2\sigma_p\right].$$

Then, a second spatial domain is defined in the order of magnitud of patchy particle size. The lower limit of this region is set at $N = 4$ and $r_b = 3/2\sigma_p$ and the upper limit is set to $N = 8$ and $r_b = 3/2\sigma_p$.

$$\lambda_2 \in \left[21/2\sigma_p, 49/2\sigma_p\right].$$

The final zone is define from the upper bound of the previous zone, to the total box length,

$$\lambda_3 \in \left[49/2\sigma_p, L\right].$$